S14 Reflection, Refraction and Diffraction | 2012 Q6

Part (a) was poorly answered. Not many candidates identified the correct positions of the anti-nodal line L· and nodal line L· by drawing a smooth curve through the points for each line. Many candidates wrongly drew straight lines for L· and L·. Only the more able ones correctly pointed out that the separation between L· and L· would increase when S·S· was reduced slightly in separation. Candidates' performance in (b) and (c) was satisfactory. Many failed to give the correct path difference at Q. In (d), the majority of candidates were able to obtain the numerical answer using the equation ·y = ·D/a but some mixed up the meaning of ·y (fringe separation) and a (slit separation).

Candidates performed well in parts (a)(i) and (ii) involving the calculations of diffraction grating. Many did not have a clear concept of percentage error in x or · and therefore failed to score marks in (a)(iii). It seems that most candidates did not understand what part (b) asked for and gave answers like using a grating with more lines per mm.

S14 Reflection, Refraction and Diffraction | 2018 Q7

Candidates performed well in (a)(i). Only a few candidates did not realise that the percentage error in measurement was being referred to and they answered 'easier to locate the positions of maxima as the dots are brighter'. Some candidates failed to find the correct values of · or mistook the order of diffraction as n = 4 in (a)(iii). Weaker ones wrongly applied ·y = ·D/a in this situation. In (b)(i), not many candidates knew that the equation ·y = ·D/a could only be applied when · · a and a · D. Some even thought that the equation was only applicable for light.

S14 Reflection, Refraction and Diffraction | 2019 Q5

This question was in general well answered. Most obtained the correct answers in (a) although a few wrongly counted the number of wavelengths in the shallow region. In (b)(i), the less able candidates struggled to find the frequency of the water wave in the deep region, which should remain unchanged (at 10 Hz). Quite a number of the candidates failed to sketch the refracted wavefronts properly in (b)(ii). Common mistakes included wavefronts bending towards instead of away from the normal or not continuing across the boundary. Part (b)(iii) was well answered except for a few who confused refraction with diffraction.

S14 Reflection, Refraction and Diffraction | 2024 Q6

Candidates' performance was satisfactory. In (a), quite a number of the candidates used the formula for double slits in their calculation, which is not applicable to diffraction grating. Part (b) was well answered.